A Complete Candidate Proof of Erdős Problem 971

Pointwise variance and a uniform three-tuple upper sieve

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Abstract

For a reduced residue class $a \pmod q$, let $p(a,q)$ be the least prime congruent to $a \pmod q$. We prove, subject only to the correctness of cited published theorems, that there are absolute constants $c, C > 0$ such that, for every sufficiently large q,

$$\#\{a \pmod q : (a,q) = 1,\ p(a,q) > (1+c)\varphi(q)\log q\} \geq C\varphi(q).$$

At the critical cutoff $X \sim \varphi(q)\log q$, the pointwise Friedlander–Goldston variance lower bound forces the second factorial moment of the prime occupancies to satisfy $C_2 \gg \varphi(q)$. A uniform three-linear-form upper sieve, followed by an explicit average of the associated singular series, gives $C_3 \ll \varphi(q)$ even for very smooth moduli. These two bounds force a positive proportion of residue classes to contain at least two primes below X. Since the total number of prime incidences is only $\varphi(q) + o(\varphi(q))$, a positive proportion of classes are empty. The prime number theorem then preserves a positive proportion of empty classes after the cutoff is enlarged to $(1+c)\varphi(q)\log q$. Every previously identified issue—centering, floors, prime powers, nonadmissible triples, coefficient uniformity, and smooth-modulus local factors—is treated explicitly.

Contents

1 Statement and external inputs

For $(a, q) = 1$, define

$$p(a, q) := \min\{p \in \mathbb{P} : p \equiv a \pmod q\}.$$

Dirichlet's theorem guarantees that this minimum exists. The problem asks whether there is an absolute $c > 0$ such that, for every sufficiently large q,

$$\#\{a \pmod q : (a, q) = 1,\ p(a, q) > (1 + c)\varphi(q)\log q\} \gg \varphi(q). \tag{1}$$

The proof uses the following two non-elementary theorems.

Theorem 1.1 (Friedlander–Goldston pointwise variance lower bound)**.** *Let*

$$V(x; q) := \sum_{a \bmod q}^{*} \left(\psi(x; q, a) - \frac{\psi(x, \chi_0)}{\varphi(q)}\right)^2.$$

For every fixed $A > 0$, uniformly for

$$\frac{x}{(\log x)^A} \le q \le x,$$

one has

$$V(x; q) \ge \left(\frac{1}{2} - o(1)\right) x \log q$$

as $x \to \infty$.

This formulation is recorded by Fiorilli from Friedlander and Goldston's theorem. The original paper uses an equivalent centering by $x/\varphi(q)$; the transfer between the two centerings is checked in section 3.

Theorem 1.2 (Uniform three-tuple upper bound)**.** *Let $a_1, a_2, b_1, b_2 \in \mathbb{Z}$, and suppose the three forms*

$$m, \qquad a_1 m + b_1, \qquad a_2 m + b_2$$

are nondegenerate and admissible. Then, uniformly in the coefficients, for all sufficiently large x,

$$\#\{p \le x : p,\ a_1 p + b_1,\ a_2 p + b_2 \text{ are prime}\} \ll \mathfrak{S}\,\frac{x}{(\log x)^3},$$

where

$$\mathfrak{S} = \prod_{\ell} \left(1 - \frac{\rho(\ell)}{\ell}\right)\left(1 - \frac{1}{\ell}\right)^{-3}$$

and $\rho(\ell)$ is the number of roots modulo ℓ of the product of the three forms. The implied constant is absolute.

For direct reproducibility one may use Dubbe's explicit Theorem 1. In the case of three forms it gives the factor 48 times the singular series and an explicit correction tending to 1. For example, once $\log x \ge 100$, its polynomial correction is less than 2, so an absolute constant 96 is sufficient.

Remark 1.3. Both inputs are unconditional. The proof below does not use GRH, Elliott–Halberstam, Hardy–Littlewood, or a lower-bound sieve for a fixed prime pair.

2 The critical cutoff

Write

$$\Phi := \varphi(q), \qquad R := \left\lfloor \frac{\Phi \log q}{q} \right\rfloor, \qquad X := qR.$$

The standard uniform estimate

$$\frac{q}{\varphi(q)} \ll \log\log q$$

implies

$$R \gg \frac{\log q}{\log\log q} \longrightarrow \infty.$$

Hence $R \geq 1$ and $X \geq q$ for all sufficiently large q. Moreover,

$$0 \leq \Phi \log q - X < q, \tag{2}$$

so

$$X = (1 + o(1))\Phi \log q. \tag{3}$$

Since $R \leq \log q$,

$$q = \frac{X}{R} \geq \frac{X}{\log q} > \frac{X}{\log X}. \tag{4}$$

Thus (X, q) lies in the range of theorem 1.1 with $A = 1$. We also have

$$\frac{X}{\Phi} = \log q + \mathrm{O}\left(\frac{q}{\Phi}\right) = \log q + \mathrm{O}(\log\log q), \tag{5}$$

$$\log X = \log q + \log R = \log q + \mathrm{O}(\log\log q), \tag{6}$$

$$\frac{X}{\log X} = (1 + o(1))\Phi. \tag{7}$$

For each reduced residue class $a \pmod q$, put

$$N_a = N_a(X) := \#\{p \leq X : p \equiv a \pmod q\}$$

and define

$$C_2 := \sum_{a \bmod q}^{*} \binom{N_a}{2}, \qquad C_3 := \sum_{a \bmod q}^{*} \binom{N_a}{3}.$$

We prove

$$C_2 \gg \Phi, \qquad C_3 \ll \Phi. \tag{8}$$

3 The second factorial moment from pointwise variance

For a reduced residue class $a \pmod q$, let

$$\psi_a := \sum_{\substack{n \leq X \\ n \equiv a\ (q)}} \Lambda(n), \qquad S := \sum_{\substack{n \leq X \\ (n,q)=1}} \Lambda(n) = \psi(X, \chi_0).$$

Set

$$V_S(X; q) := \sum_{a \bmod q}^{*} \left(\psi_a - \frac{S}{\Phi}\right)^2.$$

3.1 Centering and exact expansion

If the cited variance theorem is stated with center X/Φ, then

$$\sum_a^* \left(\psi_a - \frac{X}{\Phi}\right)^2 = V_S(X;q) + \frac{(S-X)^2}{\Phi}. \tag{9}$$

The prime number theorem gives $S = X + o(X)$, uniformly along the present sequence of q, because the omitted prime powers based on primes dividing q contribute at most

$$\sum_{p|q} \left\lfloor \frac{\log X}{\log p} \right\rfloor \log p \le \omega(q) \log X \ll (\log q)(\log X) = o(X).$$

Consequently

$$\frac{(S-X)^2}{\Phi} = o\left(\frac{X^2}{\Phi}\right) = o(X \log q),$$

by equation (5). Thus the two centerings have the same lower bound for our purpose.

Define

$$D_\Lambda := \sum_{\substack{n \le X \\ (n,q)=1}} \Lambda(n)^2$$

and

$$P_\Lambda := \sum_{\substack{m<n\le X \\ m \equiv n \ (q) \\ (mn,q)=1}} \Lambda(m)\Lambda(n).$$

Opening the square gives the exact identity

$$V_S(X;q) = D_\Lambda + 2P_\Lambda - \frac{S^2}{\Phi}. \tag{10}$$

3.2 Diagonal cancellation

The prime number theorem and partial summation give

$$\sum_{n\le X} \Lambda(n)^2 = X \log X + \mathrm{O}(X). \tag{11}$$

Removing the powers of primes dividing q costs at most

$$\sum_{p|q} \left\lfloor \frac{\log X}{\log p} \right\rfloor (\log p)^2 \le (\log X) \sum_{p|q} \log p \le (\log X) \log q = o(X \log q).$$

Therefore

$$D_\Lambda = X \log X + \mathrm{O}(X) + o(X \log q). \tag{12}$$

Also $S = X + o(X)$, so

$$\frac{S^2}{\Phi} = \frac{X^2}{\Phi} + o(X \log q).$$

Using equations (5) and (6),

$$\begin{aligned} D_\Lambda - \frac{S^2}{\Phi} &= X\left(\log X - \frac{X}{\Phi}\right) + o(X \log q) && (13) \\ &= \mathrm{O}(X \log\log q) + o(X \log q) && (14) \\ &= o(X \log q). && (15) \end{aligned}$$

Combining theorem 1.1 and equations (10) and (15),

$$P_\Lambda \geq \left(\frac{1}{4} - o(1)\right) X \log q. \tag{16}$$

3.3 Removal of proper prime powers

There are $\mathrm{O}(X^{1/2})$ proper prime powers $p^k \leq X$ with $k \geq 2$. For each such integer there are at most $R+1$ integers at most X in the same residue class modulo q, and every von Mangoldt pair weight is at most $(\log X)^2$. Thus the contribution of pairs in which at least one member is a proper prime power is

$$\mathrm{O}\Big(X^{1/2}(R+1)(\log X)^2\Big) = o(X \log q), \tag{17}$$

because $R \leq \log q \asymp \log X$. The remaining part of equation (16) is a sum over pairs of distinct primes in the same reduced residue class. Since each weight is at most $(\log X)^2$,

$$C_2 \gg \frac{X \log q}{(\log X)^2}.$$

By equations (3) and (6), there is an absolute $\alpha > 0$ such that, for all sufficiently large q,

$$\boxed{C_2 \geq \alpha\Phi.} \tag{18}$$

Integrity checkpoint 3.1 (Second-moment closure). The use of the variance theorem is pointwise in q, the centering transfer is $o(X \log q)$, the diagonal term cancels to $o(X \log q)$ at the selected cutoff, and the proper-prime-power contribution is also $o(X \log q)$. No prime-pair conjecture remains.

4 The third factorial moment from a uniform upper sieve

Every unordered triple $p_0 < p_1 < p_2 \leq X$ lying in one residue class modulo q has a unique representation

$$p_1 = p_0 + rq, \qquad p_2 = p_0 + sq, \qquad 1 \leq r < s < R.$$

Therefore

$$C_3 \leq \sum_{1 \leq r < s < R} T_q(r,s), \tag{19}$$

where

$$T_q(r,s) := \#\{p \leq X : p,\ p+rq,\ p+sq \text{ are prime}\}.$$

The enlargement of the original range $p \leq X - sq$ to $p \leq X$ only increases the count.

For a prime ℓ, let $\nu_\ell(r,s;q)$ be the number of distinct residues among

$$0,\ -rq,\ -sq \pmod{\ell}$$

and define

$$\mathfrak{S}_q(r,s) := \prod_{\ell} \left(1 - \frac{\nu_\ell(r,s;q)}{\ell}\right)\left(1 - \frac{1}{\ell}\right)^{-3}. \tag{20}$$

4.1 Admissible and nonadmissible triples

For the forms

$$p, \qquad p+rq, \qquad p+sq,$$

Dubbe's nondegeneracy determinant is

$$rs(s-r)q^3 \neq 0.$$

If the triple is admissible, theorem 1.2 gives, uniformly in q, r, s,

$$T_q(r,s) \ll \mathfrak{S}_q(r,s)\frac{X}{(\log X)^3}. \tag{21}$$

If the triple is not admissible, some prime ℓ divides at least one of the three values for every integer p. Since three residue classes cannot cover all classes modulo a prime $\ell > 3$, necessarily $\ell \leq 3$. If all three values are prime, one of them must equal ℓ. Each of the equations

$$p = \ell, \qquad p+rq = \ell, \qquad p+sq = \ell$$

has at most one integer solution. Hence

$$T_q(r,s) \leq 3 \tag{22}$$

for every nonadmissible pair (r,s).

4.2 A uniform singular-series majorant

If $\ell \mid q$, then $\nu_\ell = 1$, so the local factor is exactly

$$\left(1-\frac{1}{\ell}\right)^{-2}.$$

Thus the product of the local factors at primes dividing q is

$$\prod_{\ell \mid q}\left(1-\frac{1}{\ell}\right)^{-2} = \left(\frac{q}{\Phi}\right)^2. \tag{23}$$

Now suppose $\ell \nmid q$ and $\ell \geq 5$. If $\ell \nmid rs(s-r)$, the three roots are distinct and $\nu_\ell = 3$. The local factor is less than 1, because

$$(\ell-1)^3 - \ell^2(\ell-3) = 3\ell - 1 > 0.$$

If $\ell \mid rs(s-r)$, then $\nu_\ell \in \{1,2\}$, and the local factor is at most

$$\left(1-\frac{1}{\ell}\right)^{-2} \leq 1+\frac{4}{\ell}.$$

The last inequality follows from

$$(\ell+4)(\ell-1)^2 - \ell^3 = 2\ell^2 - 7\ell + 4 > 0 \qquad (\ell \geq 5).$$

The finitely many local possibilities at 2 and 3 contribute an absolute factor at most 16. Define

$$F(n) := \prod_{\substack{\ell \mid n \\ \ell \geq 5}}\left(1+\frac{4}{\ell}\right).$$

Then

$$\boxed{\mathfrak{S}_q(r,s) \leq 16\left(\frac{q}{\Phi}\right)^2 F(r)F(s)F(s-r).} \tag{24}$$

4.3 Average of the majorant

For every prime $\ell \geq 5$,

$$\left(1+\frac{4}{\ell}\right)^3 \leq 1+\frac{52}{\ell}.$$

Consequently

$$F(n)^3 \leq \prod_{\ell \mid n}\left(1+\frac{52}{\ell}\right) \tag{25}$$

$$= \sum_{d \mid n} \frac{\mu^2(d) 52^{\omega(d)}}{d}. \tag{26}$$

Therefore

$$\sum_{n<R} F(n)^3 \leq R \sum_{d \geq 1} \frac{\mu^2(d) 52^{\omega(d)}}{d^2} \tag{27}$$

$$= R \prod_{\ell}\left(1+\frac{52}{\ell^2}\right) \tag{28}$$

$$\ll R. \tag{29}$$

The Euler product converges absolutely. Applying Hölder's inequality over $1 \leq r < s < R$ and using equation (29) in the three variables $r, s, s-r$ gives

$$\sum_{1 \leq r<s<R} F(r) F(s) F(s-r) \ll R^2. \tag{30}$$

Combining equations (24) and (30),

$$\sum_{1 \leq r<s<R} \mathfrak{S}_q(r, s) \ll R^2\left(\frac{q}{\Phi}\right)^2. \tag{31}$$

This estimate is uniform for all moduli, including primorial and other very smooth moduli.

4.4 Completion of the third-moment bound

By equations (19), (21), (22) and (31),

$$C_3 \ll \frac{X}{(\log X)^3} R^2\left(\frac{q}{\Phi}\right)^2+R^2. \tag{32}$$

Since

$$R \leq \frac{\Phi \log q}{q},$$

we have

$$R^2\left(\frac{q}{\Phi}\right)^2 \leq(\log q)^2.$$

Hence the first term in equation (32) is

$$\ll \frac{X(\log q)^2}{(\log X)^3} \ll \Phi$$

by equations (3) and (6). Also $R^2 \leq (\log q)^2 = o(\Phi)$, using $\Phi \gg q / \log \log q$. Thus there is an absolute $\beta > 0$ such that, for all sufficiently large q,

$$\boxed{C_3 \leq \beta \Phi.} \tag{33}$$

Integrity checkpoint 4.1 (Third-moment closure). The upper sieve is uniform in the varying shifts, nonadmissible patterns contribute only $\mathrm{O}(R^2)$, and the entire smooth-modulus factor is exactly $(q/\varphi(q))^2$. The latter cancels against the number of shift pairs. No hidden $\log\log q$ loss remains.

5 Finite extraction of repeated and empty classes

Choose a fixed integer $T \geq 3$ so large that

$$\frac{3\beta}{T-1} \leq \frac{\alpha}{2}. \tag{34}$$

For every integer $k > T$,

$$\binom{k}{2} = \frac{3}{k-2}\binom{k}{3} \leq \frac{3}{T-1}\binom{k}{3}.$$

It follows from equations (33) and (34) that the contribution of the classes with $N_a > T$ to C_2 is at most $\alpha\Phi/2$. By equation (18),

$$\sum_{2 \leq N_a \leq T} \binom{N_a}{2} \geq \frac{\alpha}{2}\Phi.$$

Each summand is at most $\binom{T}{2}$, so

$$D_2 := \#\{a \pmod q : (a,q) = 1,\ N_a \geq 2\} \geq \delta\Phi, \qquad \delta := \frac{\alpha}{2\binom{T}{2}} > 0. \tag{35}$$

Let

$$M := \sum_{a \bmod q}^{*} N_a.$$

For all sufficiently large q, $X \geq q$, so every prime divisor of q is at most X. Therefore

$$M = \pi(X) - \omega(q).$$

By the prime number theorem and equation (7),

$$M = \Phi + o(\Phi). \tag{36}$$

Let $O(X)$ be the number of occupied reduced residue classes and let

$$U(X) := \Phi - O(X)$$

be the number of empty reduced classes. Every occupied class uses one prime incidence, and every class counted by D_2 uses at least one additional incidence. Hence

$$M \geq O(X) + D_2.$$

Using equations (35) and (36),

$$U(X) \geq \Phi - M + D_2 \geq (\delta - o(1))\Phi. \tag{37}$$

This is the finite implication checked by the accompanying Lean development, conditional on the two moment bounds and the total-incidence estimate.

6 Enlarging the cutoff

Let
$$Y := (1+c)\Phi \log q,$$
where $c > 0$ is fixed. From equation (3) and the prime number theorem,
$$\pi(Y) - \pi(X) = c\Phi + o(\Phi). \tag{38}$$
Indeed, $Y/X \to 1+c$, $\log X \sim \log q$, and the floor discrepancy in equation (2) contributes only $o(\Phi)$ primes.

Each newly admitted prime can fill at most one class that was empty at X. Thus
$$U(Y) \geq U(X) - \big(\pi(Y) - \pi(X)\big) \geq (\delta - c - o(1))\Phi.$$
Choose
$$c := \frac{\delta}{4}.$$
Then, for all sufficiently large q,
$$U(Y) \geq \frac{\delta}{2}\Phi. \tag{39}$$
If a reduced residue class contains no prime at most $\lfloor Y \rfloor$, its least congruent prime, being an integer, is strictly greater than Y. We have proved the following.

Theorem 6.1 (Candidate resolution of Erdős Problem 971)**.** *There exist absolute constants $c, C > 0$ and q_0 such that, for every integer $q \geq q_0$,*
$$\#\{a \pmod q : (a,q) = 1,\ p(a,q) > (1+c)\varphi(q)\log q\} \geq C\varphi(q).$$
One may take $c = \delta/4$ and $C = \delta/2$, with δ defined in equation (35).

Lean status and exact claim

The existing Lean 4 development proves the following finite implication without project-local axioms or admitted proof terms:

> If the second factorial moment has a positive linear lower bound, the third factorial moment has a linear upper bound, the total mass at the base cutoff is at most $(1 + o(1))$ times the number of classes, and the cutoff extension adds sufficiently little mass, then a positive linear number of reduced residue classes have least congruent prime beyond the enlarged cutoff.

Its strongest arithmetic specialization is named

`prime_moment_method_to_least_prime_classes`.

It also proves
$$N_a(X) = 0 \iff X < p(a,q)$$
for reduced classes, using mathlib's formalized Dirichlet theorem.

The present paper supplies the missing informal analytic instantiation:
$$C_2 \geq \alpha\Phi, \qquad C_3 \leq \beta\Phi, \qquad M = \Phi + o(\Phi), \qquad \pi(Y) - \pi(X) = c\Phi + o(\Phi).$$
However, importing these as assumptions into Lean is not the same as proving them in Lean. A genuinely complete formal verification would require substantial new mathlib developments for:

1. Chebyshev functions and their second weighted moment in the required asymptotic form;
2. the pointwise Friedlander–Goldston variance theorem;
3. the explicit or abstract three-dimensional upper-bound sieve;
4. Euler-product singular-series estimates and the Hölder average above;
5. the PNT and totient estimates assembled with all uniform quantifiers and floors.

Conclusion

The earlier bottlenecks were not instances of a remaining parity obstruction. The necessary pair lower bound is already encoded in the known pointwise variance lower bound, and the concentration problem is controlled by a standard three-tuple upper sieve. After the singular series is averaged with its exact q-local factors, the argument closes for every sufficiently large modulus.

This provides a candidate affirmative solution of Erdős Problem 971. The next step is independent review by an analytic number theorist and, if the proof survives that review, circulation as a conventional preprint or journal submission. The lack of end-to-end Lean formalization should be stated explicitly and should not be conflated with a gap in the informal analytic proof.

References

[1] T. Dubbe, *Explicit bounds for prime k-tuplets*, arXiv:2409.04261, 2024. https://arxiv.org/abs/2409.04261

[2] P. Erdős, *On some applications of Brun's method*, Acta Sci. Math. (Szeged) **13** (1950), 57–63.

[3] D. Fiorilli, *The distribution of the variance of primes in arithmetic progressions*, Int. Math. Res. Not. IMRN 2015, no. 12, 4421–4448; arXiv:1301.5663. https://arxiv.org/abs/1301.5663

[4] J. B. Friedlander and D. A. Goldston, *Variance of distribution of primes in residue classes*, Q. J. Math. **47** (1996), 313–336. https://doi.org/10.1093/qmath/47.3.313

[5] S.-K. Leung, *Moments of primes in progressions to a large modulus*, Forum Math. (2025), published online; arXiv:2402.07941. https://arxiv.org/abs/2402.07941